The Inward Bow of a Real-Rooted Polynomial

Newton's inequalities, machine-checked in Lean 4

Carles Marín*

Independent researcher

karlesmarin@gmail.com

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Abstract

I give a machine-checked, `sorry`-free proof, in Lean 4 over Mathlib, of *Newton's inequalities* (1707): if a real polynomial of degree n has all real roots, its elementary symmetric functions e_k obey

$$e_k^2 \binom{n}{k-1}\binom{n}{k+1} \;\geq\; e_{k-1}\, e_{k+1} \binom{n}{k}^2, \qquad 1 \leq k \leq n-1,$$

equivalently the normalized means $p_k = e_k/\binom{n}{k}$ are log-concave. This is the classical bridge *real-rooted* $\Rightarrow$ *log-concave*, the elementary base beneath a large body of modern combinatorics—the matching polynomial, the Heilmann–Lieb theorem, the Hodge-theoretic proofs of matroid log-concavity. Mathlib already carries the elementary symmetric functions (`Multiset.esymm`) and the Vieta relations; it did not carry the inequality, and to the best of my knowledge no proof assistant did. The proof is the classical reduction made fully rigorous: real-rootedness survives differentiation (Rolle) and reversal ($x \mapsto 1/x$), so differentiating $i-1$ times, reversing, and differentiating again collapses the k-th instance to a real-rooted *quadratic*, whose discriminant $b^2 - 4ac \geq 0$ *is* the inequality once positive factorials are cleared. I am exact about the boundary: the mathematics is entirely classical (Newton, Maclaurin, Sylvester); the contribution is the formalization—together with four reusable building blocks (real-rootedness under differentiation, iterated differentiation, and reversal; the coefficient-form discriminant)—and the fact that the real-rooted⇒log-concave engine, used informally throughout this line of work, now stands certified. The headline theorem and its blocks depend only on `propext`, `Classical.choice`, `Quot.sound`.

How to read this paper. It is the story of four questions, each opening a section and handing the reader the next. The first asks what reality of the roots forces on the coefficients (Q1); the second, how to turn that into something a machine finishes in one line (Q2); the third, what the formalization caught that the textbook glides over (Q3); the fourth, what falls out once the inequality is certified (Q4). If you read only one thing, read Theorem 1 and Figure 2; the Lean inventory is Table 1; the honest limits are in Section 6; reproduction is three commands.

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author's responsibility. The methodology and the exact boundary of the contribution are in Section 6.

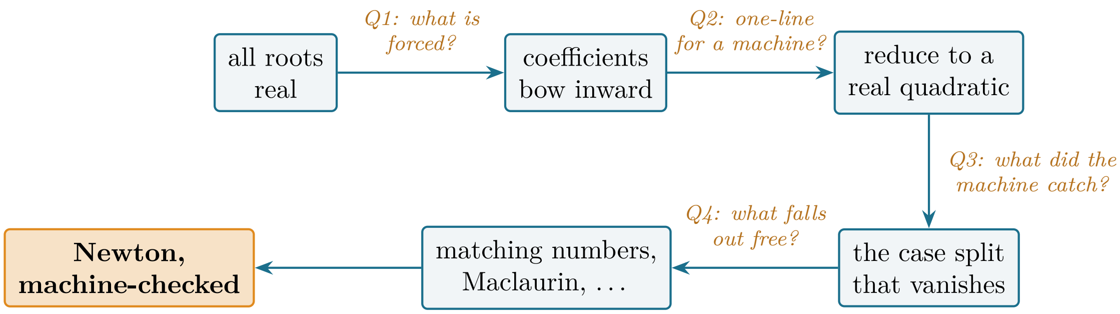

Figure 1: The reader's map. From a hypothesis on the roots to a certified inequality and its corollaries; each arrow is the question its section answers. Every node is a block of `sorry`-free Lean.

1 Q1: what does reality of the roots force on the coefficients?

Tell me nothing about a polynomial except that all of its roots are real—not where they are, not whether they repeat—and I can already constrain its coefficients. That such a thing is possible is mildly surprising; that the constraint is sharp, classical, and still doing heavy lifting in today's combinatorics is the reason for this paper.

History bead. The inequalities are Newton's, stated without proof in the *Arithmetica Universalis* (1707) [1]. Maclaurin (1729) deduced from them the chain of mean inequalities that bears his name [2]; a rigorous proof came only in the nineteenth century (Sylvester), and the modern textbook account is Hardy–Littlewood–Pólya [3]. The mechanism—real-rootedness inherited by the derivative (Rolle) and by the reversal of the polynomial—is the one formalized here.

Write a real-rooted polynomial through its roots, $p(x) = \sum_{k=0}^{n} (-1)^k e_k \, x^{n-k}$, where e_k is the k-th *elementary symmetric function* of the roots, and form the *normalized means* $p_k = e_k / \binom{n}{k}$. Newton's claim is that the means are *log-concave*—each one at least the geometric mean of its neighbours. Three centuries on, this is the cheapest instance of a pattern that organizes a large part of modern combinatorics: to prove a sequence log-concave (hence unimodal, hence well-behaved), exhibit it as the coefficients of a *real-rooted* polynomial, and let Newton finish. The strategy underlies Stanley's survey of log-concave sequences [4], Brändén's theory of stable polynomials [5], and—when no real-rooted polynomial is at hand—the deeper Hodge-theoretic proof of Adiprasito–Huh–Katz that the characteristic polynomial of a matroid is log-concave [6]. Newton's inequalities are the elementary, real-rooted floor of that whole building.

Theorem 1 (Newton's inequalities, in Lean 4; `Newton.newton_inequality`)**.** *For a real-rooted real polynomial p of degree n and every $1 \le k \le n-1$,*

$$\boxed{e_k^2 \binom{n}{k-1}\binom{n}{k+1} \;\ge\; e_{k-1}\, e_{k+1} \binom{n}{k}^2 .}$$

Equivalently (`newton_logConcave`) the normalized means $p_k = e_k/\binom{n}{k}$ are log-concave, $p_{k-1}\, p_{k+1} \le p_k^2$. Both are `sorry`-free; `#print axioms` on each reports only `propext`, `Classical.choice`, `Quot.sound`.

Hand-checkable case. Take $p(x) = (x-1)(x-2)(x-4) = x^3 - 7x^2 + 14x - 8$, so $n = 3$ and $(e_0, e_1, e_2, e_3) = (1, 7, 14, 8)$ (Figure 2, left). At $k = 1$ the theorem reads $e_1^2 \binom{3}{0}\binom{3}{2} = 147 \ge e_0 e_2 \binom{3}{1}^2 = 126$; at $k = 2$, $588 \ge 504$. The all-roots-equal polynomial $(x-1)^4$ *saturates* every inequality (the

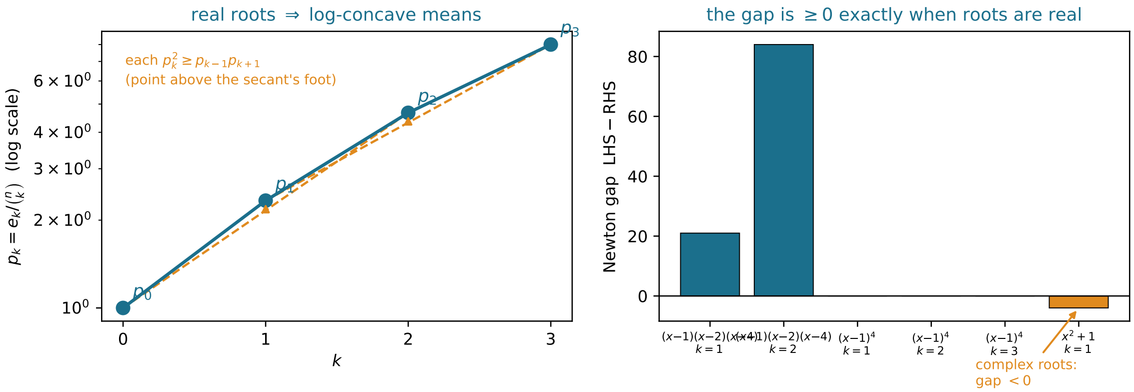

Figure 2: *Left:* the normalized means of $(x-1)(x-2)(x-4)$, plotted on a log axis; each interior mean sits above the geometric mean of its neighbours (the amber arrows), so the sequence is concave—it "bows inward." *Right:* the Newton gap $e_k^2\binom{n}{k-1}\binom{n}{k+1} - e_{k-1}e_{k+1}\binom{n}{k}^2$ at every interior k: positive for the real-rooted cubic, exactly zero for $(x-1)^4$ (the sharp case), and *negative* for x^2+1. The generating script asserts every plotted value against exact integer arithmetic.

sharp boundary), and x^2+1—whose roots are not real—*fails* it ($0 \ngeq 4$ at $k=1$), a reminder that the hypothesis is load-bearing, not decorative (Figure 2, right).

One feature of the statement repays a second look:

> *why should the gap between real and complex roots surface precisely in the elementary symmetric functions, and precisely as log-concavity?*

Because the e_k *are* the coefficients of the polynomial, read through Vieta, and real-rootedness is a property that propagates downward through the entire tower of derivatives (Rolle) and survives reversing the polynomial. Those two moves let one isolate any three consecutive coefficients as the coefficients of a real-rooted quadratic—and for a quadratic, "real-rooted" is literally "discriminant ≥ 0." That is Newton's inequality, and the next section is how a machine walks the same path. *Can the textbook gesture become a one-line check?*

2 Q2: how do you make it a one-line check for a machine?

History bead. The differentiation–reversal proof is the classical one (Sylvester; the account in [3]): grind the index down to a quadratic using operations that preserve reality of roots, then invoke the trivial $n=2$ case. The whole question is whether "preserves real roots" and "reduce to a quadratic" survive being made fully formal.

The objects. I work over $\mathbb{R}$. A polynomial is *real-rooted* (`RealRooted`) when it splits into linear factors over $\mathbb{R}$—Mathlib's intrinsic `Polynomial.Splits`; equivalently it has $\deg p$ real roots counted with multiplicity. For the roots multiset $s = p$.`roots`, the elementary symmetric function is e_k = `Multiset.esymm` $s\,k$, and $\binom{n}{k}$ = `Nat.choose`. Two pieces of Mathlib do the bridging: `coeff_eq_esymm_roots_of_card` is Vieta, p.`coeff` $j = p$.`leadingCoeff` $\cdot\,(-1)^{n-j}\,e_{n-j}$ (so I may pass between the symmetric-function statement and a statement about coefficients), and `card_roots_le_derivative` is Rolle for polynomials. Mathlib had both halves; it did not have the inequality on them.

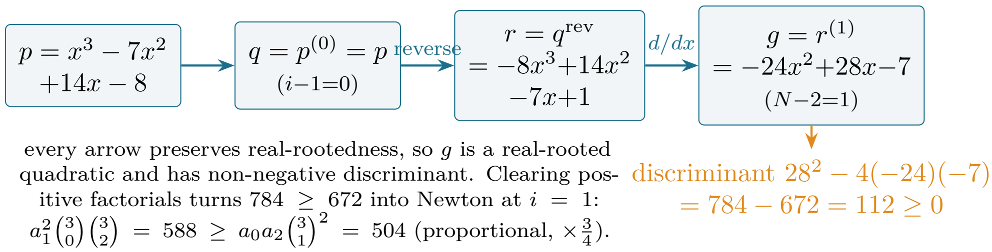

Figure 3: The differentiate–reverse reduction, worked on the running cubic at $i = 1$ (relating $a_0, a_1, a_2 = -8, 14, -7$). The final object is a quadratic; a real-rooted quadratic has non-negative discriminant; the factorials the proof cancels are the proportionality constant.

The two stones. Exactly two operations preserve real-rootedness and that is all the proof needs. *Differentiation:* between consecutive real roots of p lies a real root of p' (Rolle), so p' has $\geq \deg p - 1$ real roots, which is all it can have (`RealRooted.derivative`, iterated to `RealRooted.iterate_derivative`). *Reversal:* p^{rev} is, up to a nonzero factor, p evaluated at $1/x$, so its roots are the reciprocals of the nonzero roots of p, still real (`RealRooted.reverse`). With these, the reduction is short.

The reduction (recipe)

To prove the inequality at index i for a real-rooted p of degree n (set $N = n - i + 1$): (1) differentiate $i - 1$ times: $q = p^{(i-1)}$, real-rooted of degree N; (2) reverse: $r = q^{\mathrm{rev}}$, real-rooted; (3) differentiate $N-2$ times: $g = r^{(N-2)}$, real-rooted of degree ≤ 2. The three low coefficients of g are *factorial multiples* of a_{i-1}, a_i, a_{i+1} (writing $a_j = p.\texttt{coeff}\,j$), and the base case $g.\texttt{coeff}\,1^2 \geq 4\,g.\texttt{coeff}\,2 \cdot g.\texttt{coeff}\,0$ is, after clearing those positive factorials, Newton's inequality at i.

The base case, in coefficients. The $n = 2$ heart is `realRooted_discrim_coeff`: a real-rooted q with $\deg q \leq 2$ satisfies $q.\texttt{coeff}\,1^2 \geq 4\,q.\texttt{coeff}\,2 \cdot q.\texttt{coeff}\,0$. Evaluate q at one of its roots x and read off $b^2 - 4ac = (2ax + b)^2 - 4a(ax^2 + bx + c) = (2ax + b)^2 \geq 0$. Stating it on *coefficients*, with the weak hypothesis $\deg \leq 2$, is the choice that makes the whole proof clean (Section 3).

The bookkeeping. The one technical part is computing the three low coefficients of g. Mathlib's `coeff_iterate_derivative` gives $(p^{(k)}).\texttt{coeff}\,m = (m+k)^{\underline{k}}\,p.\texttt{coeff}(m+k)$ (a falling factorial), and `coeff_reverse` gives $q^{\mathrm{rev}}.\texttt{coeff}\,m = q.\texttt{coeff}(\mathrm{revAt}\,(\deg q)\,m)$. Composing derivative–reverse–derivative needs the *exact* degree of each intermediate, which over $\mathbb{R}$ (characteristic zero, an `IsAddTorsionFree` ring) is supplied by `degree_derivative_eq`: differentiating a degree-$d{>}0$ polynomial drops the degree by exactly one. With degrees pinned, the discriminant inequality, after rewriting each falling factorial through `Nat.factorial_mul_descFactorial`, becomes

$$\boxed{\big((N-1)!\,i!\big)^2\,a_i^2 \;\geq\; N!\,(N-2)!\,(i-1)!\,(i+1)!\,a_{i-1}a_{i+1},}$$

and the binomial identity $\binom{n}{k}\,k!\,(n-k)! = n!$ (`Nat.choose_mul_factorial_mul_factorial`) clears these positive factors to match Theorem 1—no inequality reasoning, only positivity. The symmetric-function form then follows from the coefficient form (`newton_inequality_coeff`) by Vieta, the signs $(-1)^{k-1}(-1)^{k+1} = 1$ cancelling and the binomials reindexing by $\binom{n}{k} = \binom{n}{n-k}$ to the running index $i = n - k$.

So the answer to Q2 is yes: the inequality reduces, in three reality-preserving moves, to a single discriminant a machine closes by `sq_nonneg`. *And at the one place the textbook waves its hand, the machine made me earn it.*

3 Q3: what did the machine catch that the textbook glides over?

History bead. "Differentiate, reverse, reduce to a quadratic" is how the classical proof runs, and the classical proof quietly assumes the reversed polynomial keeps its degree. A proof assistant does not let the assumption pass unexamined—and examining it removed a case split, not added one.

The reversal step preserves *degree* only when the reversed polynomial's leading coefficient is nonzero, which fails exactly when $a_{i-1} = 0$. The textbook reflex is a case split: handle $a_{i-1} = 0$ separately. I avoid it entirely, and the reason is the shape of the base case.

What a proof assistant earns its keep on

Phrasing the base case on *coefficients* with the weak hypothesis $\deg \le 2$ (not $= 2$) makes it return a true statement even when the quadratic degenerates: if $g.\texttt{coeff}\,2 = 0$ the inequality reads $g.\texttt{coeff}\,1^2 \ge 0$, which is `sq_nonneg`. Traced back, when $a_{i-1} = 0$ Newton's right-hand side is $a_{i-1}a_{i+1}\binom{n}{i}^2 = 0$ and the left-hand side is a square times non-negative binomials—so the inequality holds for a reason unrelated to the quadratic. *The general machinery and the degenerate case are discharged by the same line.* The transferable lesson: choosing the weakest honest hypothesis for a base case often dissolves the edge cases a sharper statement would force you to chase.

This is the whole of what the formalization added beyond transcription, and it is honestly small—not a theorem but a discipline. It is also exactly the kind of step that justifies machine-checking a result everyone already believes: the place where "without loss of generality" turns out to cost nothing, but only once you have looked. *With Newton certified, what does the rest of this line of work get for free?*

4 Q4: with Newton certified, what falls out for free?

History bead. The implication "real-rooted ⇒ log-concave coefficients" is the workhorse of the field: it is why the matching numbers of a graph are log-concave (via Heilmann–Lieb [8]), why the coefficients of a claw-free graph's independence polynomial are, and the easy half of the results that, for matroids, needed Hodge theory because no real-rooted polynomial was available [6].

The coefficient-form lemma `newton_inequality_coeff` is stated precisely so that feeding it any real-rooted polynomial yields log-concavity of that polynomial's coefficients, with no further work:

$$\boxed{\text{real-rooted} \implies \text{log-concave coefficients.}}$$

The matching polynomial of a graph is real-rooted (Heilmann–Lieb, the previous paper in this line [8]); composing it with the lemma proved here gives log-concavity of the matching numbers as an immediate corollary, rather than the textbook gesture it has been throughout this series. The same one line applies to any future real-rooted polynomial a Lean development produces.

Two roads open directly upward (Table 1 lists what is built; these are what is not yet):

- **Maclaurin's mean inequalities** $p_1 \geq p_2^{1/2} \geq \cdots \geq p_n^{1/n}$, Newton's immediate historical consequence—now a concrete next target rather than a citation.
- **The equality cases**: that equality at an interior k forces all roots equal, the classical sharp companion, visible as the $(x-1)^4$ row of Figure 2 but not yet proved.

Here is a question worth carrying away:

> *why should two operations as different as* differentiating *and* reciprocating the roots *both preserve reality of roots—and why is that pair exactly enough to expose every coefficient window?*

Because both are shadows of one fact: the real-rooted polynomials form a closed world (the several-variable home is Brändén's stable polynomials [5]). Differentiation is the simplest linear operator preserving it, reversal the simplest involution; between them they act on coefficient windows transitively enough to surface any three consecutive coefficients as a quadratic. Newton's inequalities are the first visible consequence of that closure—and the reduction in this paper is, in miniature, how every later real-rootedness argument borrows its strength.

5 The building blocks

The headline rests on four reusable pieces—the genuinely portable output—each `sorry`-free with the standard three axioms (Table 1).

Lemma 1 (`RealRooted.derivative`, `RealRooted.iterate_derivative`)**.** *If p is real-rooted then so is p', and hence every iterated derivative $p^{(k)}$.*

Rolle's theorem for polynomials (`card_roots_le_derivative`) squeezed against the degree drop: p' already has $\deg p - 1$ real roots, which is the most it can have.

Lemma 2 (`RealRooted.reverse`)**.** *If p is real-rooted then so is its reversal p^{rev}.*

By factoring p into monic linears and reversing each; the constant-term hypothesis the textbook attaches is *unnecessary* and is dropped—reversal merely discards trailing zeros, and the remaining product of ≤ 1-degree factors still splits.

Lemma 3 (`realRooted_discrim_coeff`)**.** *A real-rooted q with $\deg q \leq 2$ satisfies $q.\texttt{coeff}\,1^2 \geq 4\,q.\texttt{coeff}\,2 \cdot q.\texttt{coeff}\,0$.*

The base case of the reduction, in coefficient form; the source of the dissolved case split (Section 3).

Lean name	statement	role
`newton_inequality`	$e_k^2\binom{n}{k-1}\binom{n}{k+1} \geq e_{k-1}e_{k+1}\binom{n}{k}^2$	Theorem 1
`newton_logConcave`	$p_{k-1}p_{k+1} \leq p_k^2$ (normalized means)	corollary
`newton_inequality_coeff`	$a_i^2\binom{n}{i-1}\binom{n}{i+1} \geq a_{i-1}a_{i+1}\binom{n}{i}^2$	coefficient form
`realRooted_discrim_coeff`	real-rooted, deg $\leq 2 \Rightarrow a_1^2 \geq 4a_2a_0$	base case
`RealRooted.derivative`	p real-rooted $\Rightarrow p'$ real-rooted	block (Rolle)
`RealRooted.iterate_derivative`	p real-rooted $\Rightarrow p^{(k)}$ real-rooted	block
`RealRooted.reverse`	p real-rooted $\Rightarrow p^{\mathrm{rev}}$ real-rooted	block

Table 1: The load-bearing results, all `sorry`-free in `Newton.lean` of the repository (346 lines, on `RealStable.lean` and Mathlib). Axioms of every entry: `propext`, `Classical.choice`, `Quot.sound` only; no `native_decide`, no custom axiom.

6 The boundary of the contribution

Everything here that is mathematics is classical: Newton (1707), Maclaurin (1729), Sylvester's proof, the textbook treatment in Hardy–Littlewood–Pólya [3]. I claim no new identity and no new inequality. The contributions are three: the *formalization* (to my knowledge the first machine-checked proof of Newton's inequalities in any proof assistant); the *reusable infrastructure*—real-rootedness under differentiation, iterated differentiation, and reversal, and the coefficient-form discriminant—which is exactly what any further real-rootedness or stability development in Mathlib will want; and the *proof-engineering record* of Section 3, the weak-hypothesis base case that dissolves a case split.

Scope, stated plainly. I prove the inequalities *over* $\mathbb{R}$—the natural home of real-rootedness, and what the reduction uses—not over an abstract real-closed or ordered field, though nothing in the argument resists it. I prove the inequality and its log-concave corollary, *not* the equality/strictness analysis (equality forces all roots equal), and not Maclaurin's mean chain: those are mapped (Section 4), not built.

Prior formalizations. The priority claim rests on a documented search, not a feeling. For Newton's inequalities and the elementary symmetric functions I searched (June 2026): Mathlib (Loogle and `grep`—`Multiset.esymm` and the Vieta relations exist; *Maclaurin* and `esymm`-squared inequalities do not), the wider Lean community repositories, the Isabelle Archive of Formal Proofs, and the Coq/Rocq ecosystem. None contains Newton's inequalities. As always, such a claim is made to the best of my knowledge.

On methodology. The formalization was done with an AI assistant (Claude, Anthropic), under the discipline of the companion papers: the reduction's arithmetic verified numerically before formalizing (`figures/newton_fig1_bow.py` re-checks every instance), and every named theorem verified by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`. Correctness does not rest on the assistant: the Lean kernel checks every proof. The careful accounting—the scope over $\mathbb{R}$ stated as plainly as the result, the equality cases left explicitly open—is itself part of what the method, at its best, should produce. The mathematics and the claims are the author's responsibility.

7 Open questions

Newton's inequalities are now a theorem in the library. Standing on them, five questions worth the next effort.

1. *Maclaurin, mechanically.* Maclaurin's mean chain follows from Newton by a short argument.

What is the smallest Lean bridge that turns the inequality proved here into the full chain $p_1 \geq p_2^{1/2} \geq \cdots \geq p_n^{1/n}$?

2. *The sharp case.* Equality at an interior k forces all roots equal. Formalizing that needs the strict version of Rolle and of the discriminant; what is the cleanest route, and does the weak-hypothesis base case still help or now get in the way?

3. *Matching numbers as a corollary.* Composing `newton_inequality_coeff` with the Heilmann–Lieb real-rootedness of the matching polynomial [8] should print log-concavity of the matching numbers in a few lines. How few?

4. *The closure made explicit.* Differentiation and reversal both preserve real-rootedness. What is the largest natural family of coefficient-window operations that do, and is "enough to reach any quadratic window" a theorem one could state and check?

5. *Beyond real roots.* The Hodge-theoretic log-concavity results [6] hold where no real-rooted polynomial exists. Newton is the real-rooted floor; what is the smallest formalized fragment of the matroid story that would sit one storey above it?

Reproducing

```
git clone https://github.com/karlesmarin/godsil-gutman-lean
cd godsil-gutman-lean && lake exe cache get
lake env lean Newton.lean
```

Lean `v4.30.0-rc2`, Mathlib pin `701fb6e9`. The theorems are `Newton.newton_inequality`, `newton_logConcave` and `newton_inequality_coeff` in `Newton.lean`; their supports are in Table 1. Each is checked by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`. The figure regenerates with `python figures/newton_fig1_bow.py`, which asserts its content against exact arithmetic before drawing.

References

[1] I. Newton, *Arithmetica Universalis* (1707); the inequalities among the coefficients of a polynomial with real roots are stated there without proof.

[2] C. Maclaurin, A second letter ... concerning the roots of equations, with the demonstration of other rules in algebra, *Phil. Trans. Roy. Soc. London* **36** (1729) 59–96.

[3] G. H. Hardy, J. E. Littlewood, G. Pólya, *Inequalities*, 2nd ed., Cambridge Univ. Press, 1952 (Newton's and Maclaurin's inequalities, §2.22).

[4] R. P. Stanley, Log-concave and unimodal sequences in algebra, combinatorics, and geometry, *Ann. New York Acad. Sci.* **576** (1989) 500–535.

[5] P. Brändén, Unimodality, log-concavity, real-rootedness and beyond, in *Handbook of Enumerative Combinatorics*, CRC Press, 2015, 437–483.

[6] K. Adiprasito, J. Huh, E. Katz, Hodge theory for combinatorial geometries, *Ann. of Math.* **188** (2018) 381–452.

[7] C. Marín, *Random Signs into Matchings: A Godsil–Gutman Identity, Formalized in Lean 4*, Zenodo, 2026. `doi:10.5281/zenodo.20517350`.

[8] C. Marín, *Unfolding a Graph into a Tree: A Machine-Checked Proof of the Heilmann–Lieb Theorem in Lean 4*, Zenodo, 2026. `https://github.com/karlesmarin/godsil-gutman-lean`.

[9] The mathlib Community, The Lean mathematical library, in *CPP 2020*, ACM, 2020, 367–381.